

Nanostructured polymers: Macromolecular engineering

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Nanostructured polymers: Macromolecular engineering

Outline

- Chapter 1: Introduction to Nanostructured polymers
- Chapter 2: Anionic living polymerization
- Chapter 3: Controlled radical polymerization: From conventional free radical polymerization to reversible-deactivation radical polymerization (RDRP)
- Chapter 4: Controlled radical polymerization: The NMP, ATRP, and RAFT processes
- Chapter 5: Ring-opening metathesis polymerization (ROMP)
- Chapter 6: Macromolecular engineering: Strategies and methods for the synthesis of nanostructured polymers (functional polymers, block and graft copolymers, hyperbranched polymers, dendrimers...)
- Chapter 7: Nanostructured polymers - case studies: Block and graft copolymers.

Nanostructured polymers: Macromolecular engineering

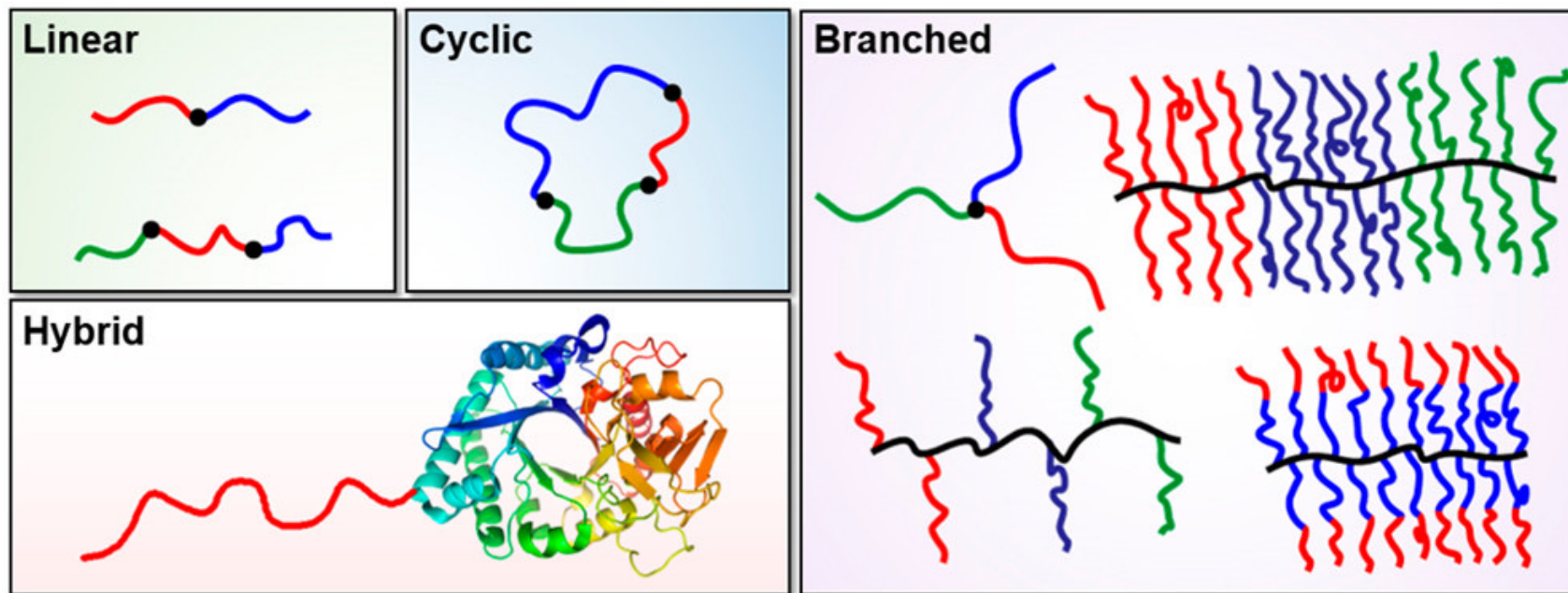
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Polymers as building blocks for Nanomaterials

From **Macromolecules** to **Nanomaterials**:

Macromolecular engineering based on **modern synthetic chemistry** methods offers possibilities to design polymers with specific composition, dimensions, and topology.



Chapter 2

Anionic Living Polymerization

- Introduction
- Chain-growth polymerization: General scheme
- Anionic polymerization: General scheme
- Living anionic polymerization: Definition
- Examples
- Monomers
- Consequences of termination and transfer-free polymerization
- Conclusions & Definitions
- Summary

Introduction

Polymers can be prepared *via* different **synthetic methods**:

- **Chain-growth polymerization**: anionic, cationic, radical, or coordination
- Step-growth polymerization: **polycondensation**
- **Post-polymerization modification** (PPM or chemical modification)

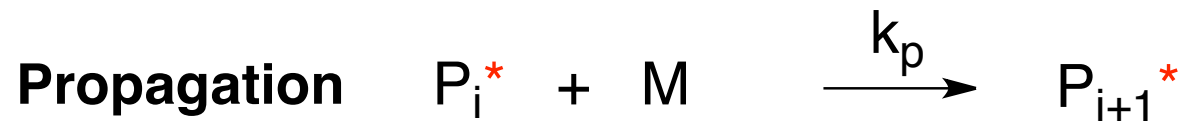
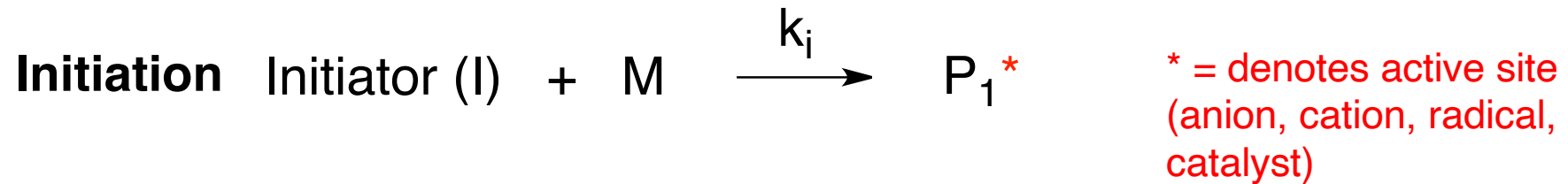
... Or provided by Nature:

- **Biopolymers** (peptides, nucleic acids, polysaccharides, etc.)



*We will only deal here with **chain polymerization reactions**.*

Chain-growth polymerization: General scheme



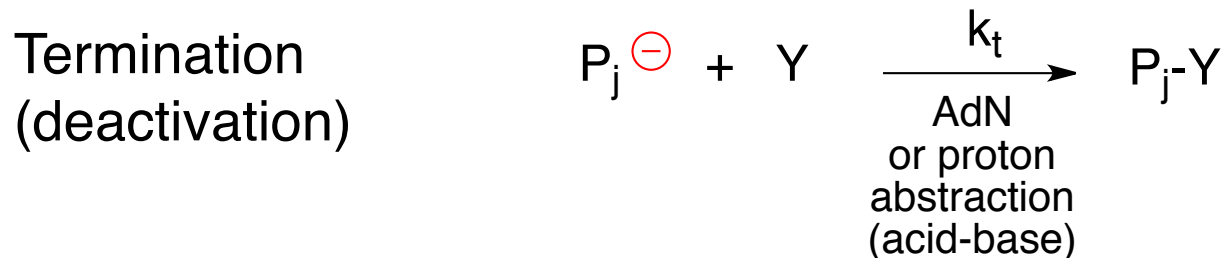
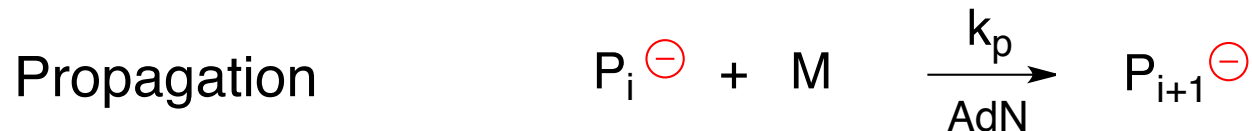
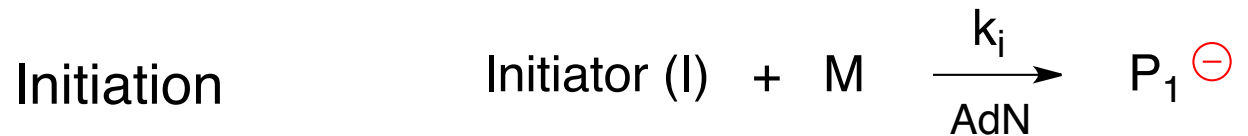
These two steps do not always occur during the process!

T-Z = transfer agent: any (macro)molecule present in the reaction medium: monomer, initiator, solvent, polymer...

T^{*} should restart a new active chain by reaction with the monomer (M) with a rate similar to that of initiation.

Anionic polymerization

General scheme: *via* nucleophilic addition (AdN) mechanism

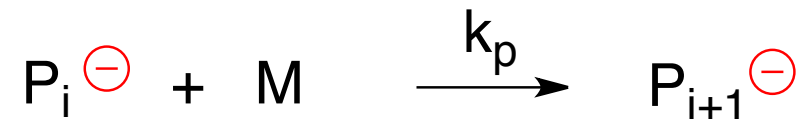


Most frequently, the **propagating anionic active site** is a carbanion or oxanion, *i.e.*, strongly nucleophilic and basic species, very reactive with electrophilic and acid impurities (such as water...), so anhydrous conditions are required!

Anionic living polymerization

Anionic polymerization **with no termination or transfer** was named anionic **living polymerization** by Szwarc (1956).

The **propagating anionic active site** remains active after monomer is depleted (block copolymers can thus be prepared).



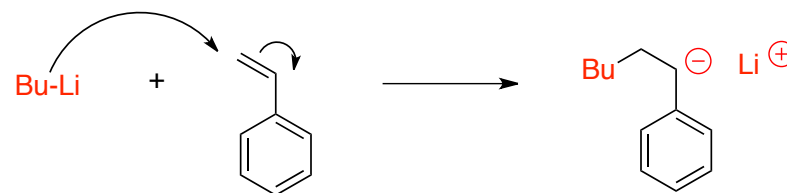
Termination is usually done deliberately (by the manipulator) by reaction with an acid ($\text{Y} = \text{H}_2\text{O}$, ROH) or electrophilic species (which allows the terminal functionalization of the chains):



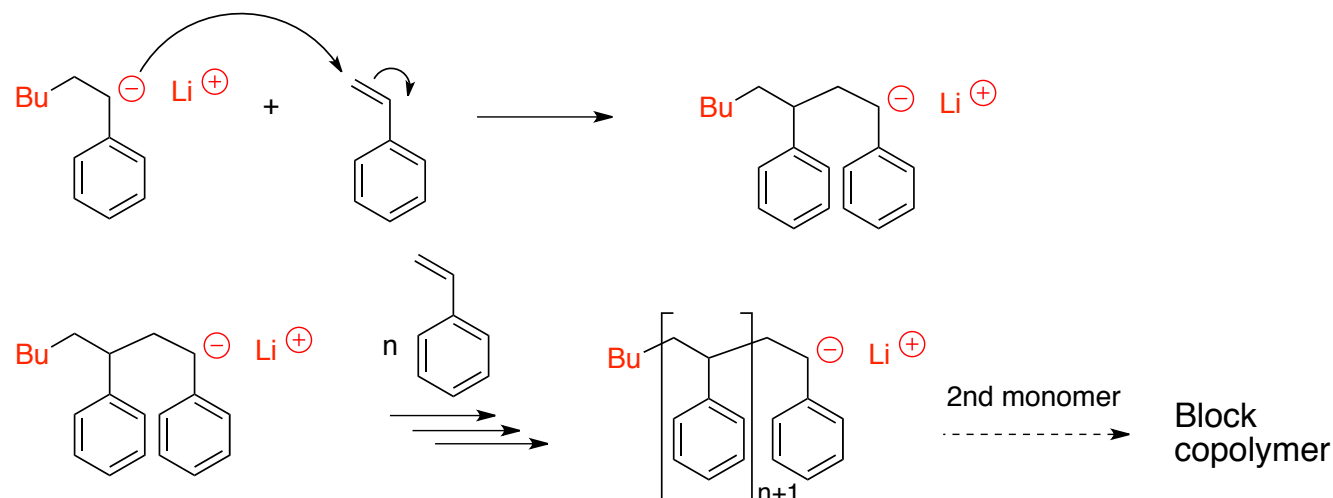
Anionic living polymerization

Example (1): Anionic polymerization of styrene using an alkyllithium (*s*-BuLi) initiator

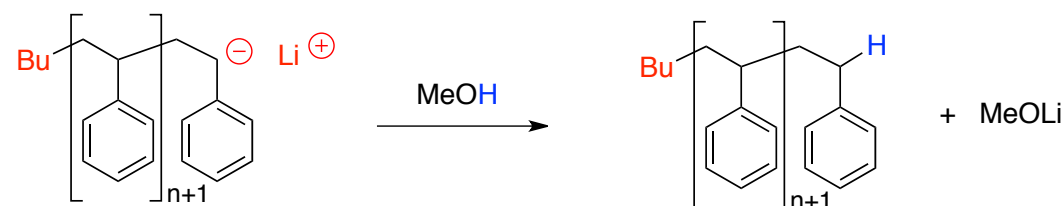
Initiation



Propagation



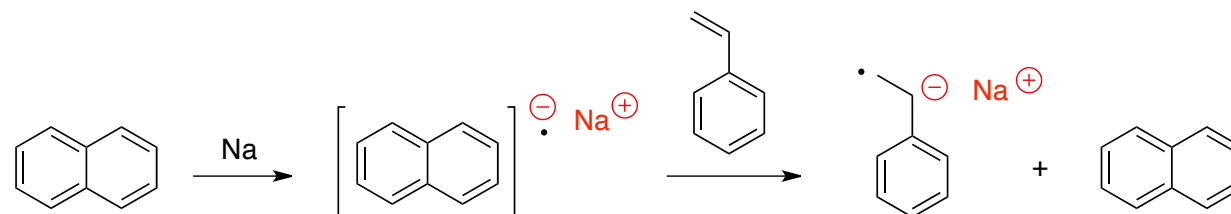
Termination
by deactivation
using MeOH



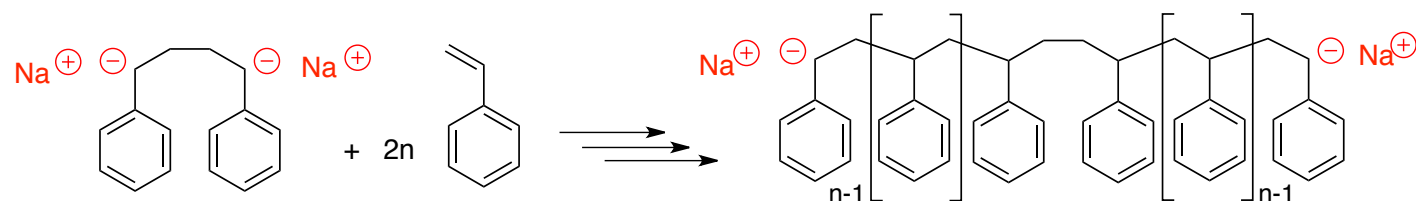
Anionic living polymerization

Example (2): Anionic polymerization of styrene using the naphthalene / sodium system

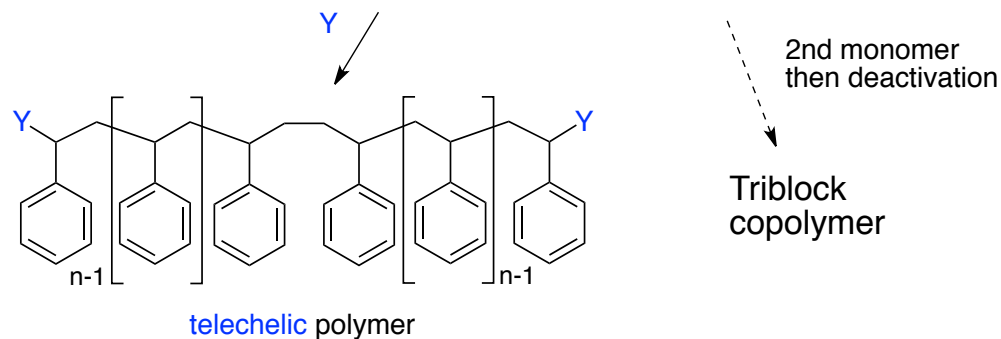
Initiation



Propagation



Termination
by deactivation using
an electrophilic
reagent Y



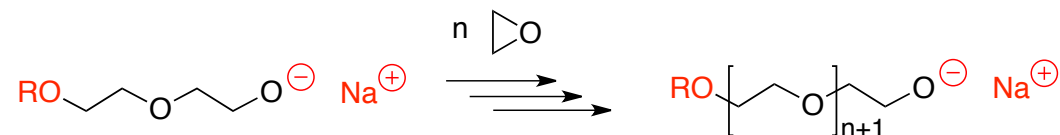
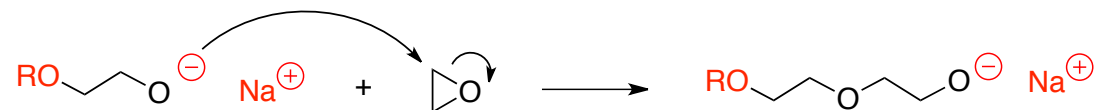
Anionic living polymerization

Example (3): Anionic ring-opening polymerization of ethylene oxide

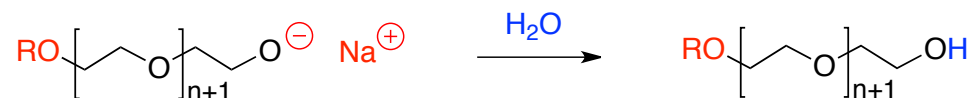
Initiation



Propagation



Termination
by deactivation
using, *e. g.*, water

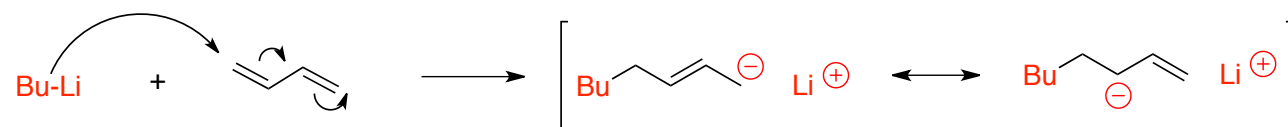


Note: Reaction works with Na, K, ...
but No polymerization occurs with the Li counter-ion!

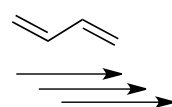
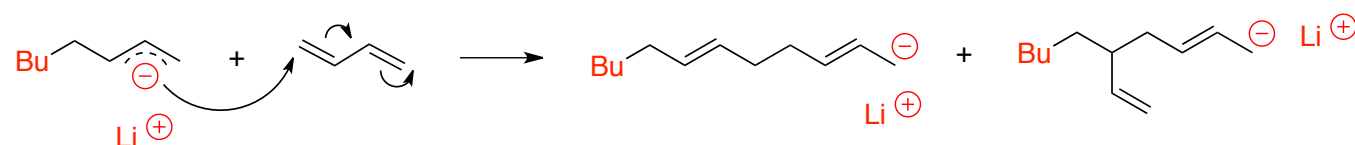
Anionic living polymerization

Example (4): Anionic polymerization of dienes

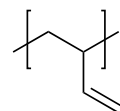
Initiation



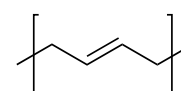
Propagation



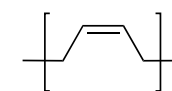
Polybutadiene (PBu)



-1,2 unit

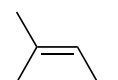


-1,4 *trans* unit

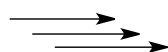


-1,4 *cis* unit

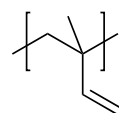
Polymerization of isoprene:



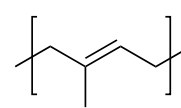
Isoprene



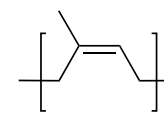
Polyisoprene (PI)



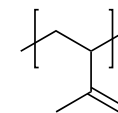
-1,2 unit



-1,4 *trans* unit



-1,4 *cis* unit



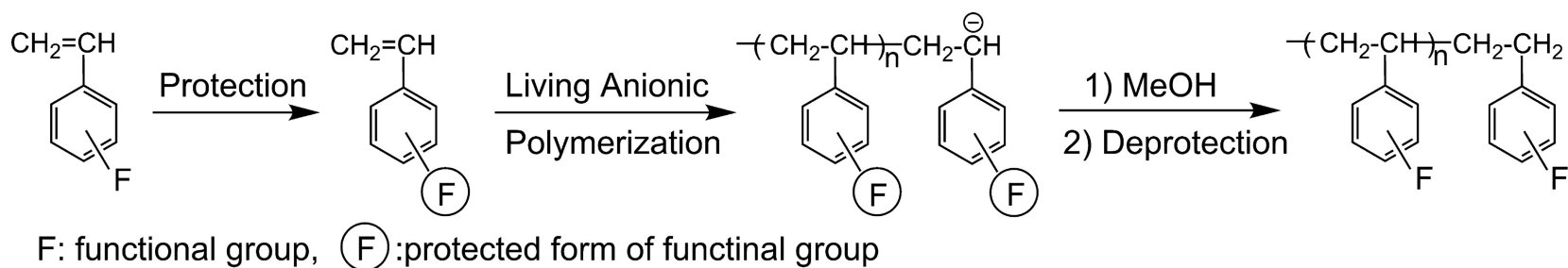
-3,4 unit



The conditions (solvent, counterion, ...) play a determining role on the type of units and the stereochemistry of the polymer formed.

Anionic living polymerization: Monomers

- The initiators and active centers are generally strong **nucleophiles** and/or **bases** that easily react with **electrophilic** and/or **acidic** (protic) functionalities.
- Accordingly, functional groups having active hydrogen(s) and carbonyl groups, such as OH, SH, SiOH, NH₂, C≡CH, CHO, COR, and COOH, are not compatible with anionic initiators and chain-end anions, thus rendering living anionic polymerization of substituted monomers with such functionalities difficult or impossible.
- To overcome this problem, a **protection strategy** can be used, as illustrated into living anionic polymerization of functional styrenes:



Consequences of Termination and Transfer-Free Polymerization (1)

Assuming a fast initiation step, the rate of polymerization is given by:

$$R_p = -\frac{d[M]}{dt} = k_p \cdot [P^*] \cdot [M]$$

where $[M]$ is the monomer concentration, k_p is the rate constant of propagation, and $[P^*]$ is the concentration of active chain ends.

In the absence of termination, $[P^*]$ is constant, and the product $k_p[P^*] = k_{app}$ can be regarded as an apparent first-order rate constant.

Introducing monomer conversion, $x_p = ([M]_0 - [M]_t)/[M]_0$, integration of rate equation leads to:

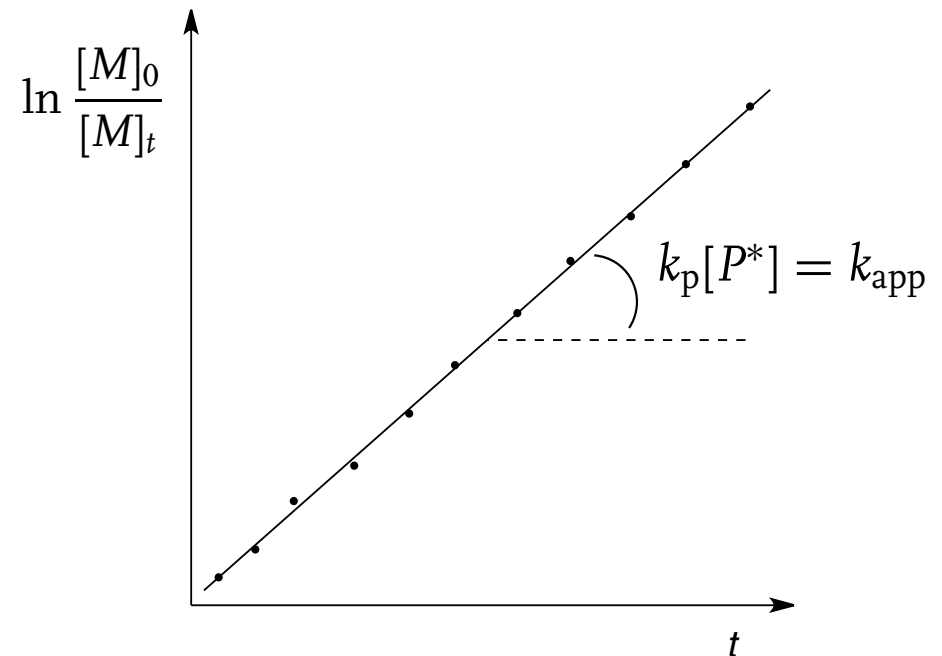
$$\ln \frac{[M]_0}{[M]_t} = -\ln(1 - x_p) = k_p \cdot [P^*] \cdot t = k_{app}t$$

Consequences of Termination and Transfer-Free Polymerization (2)

$$\ln \frac{[M]_0}{[M]_t} = -\ln(1 - x_p) = k_p \cdot [P^*] \cdot t = k_{app} t$$

The linearity of the plot of such a first-order time-conversion relation indicates that the **active center concentration remains constant** throughout the polymerization.

In case of termination, $[P^*]$ depletes and thus the slope of the first-order plot decreases.



It must be noted that this plot does not give evidence for the absence of transfer, since in this case the concentration of active chain ends remains constant.

Consequences of Termination and Transfer-Free Polymerization (3)

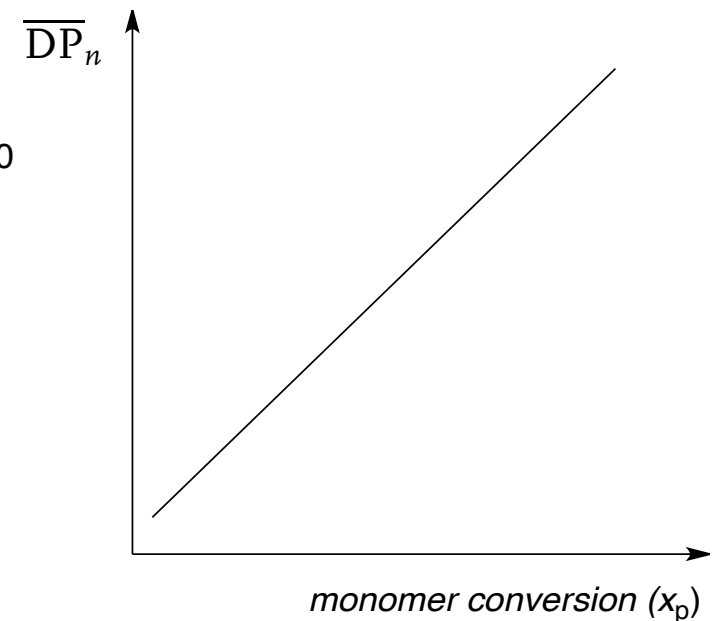
The **absence of transfer** can be demonstrated by the linearity of a plot of the number-average degree of polymerization, $\overline{DP}_n$, vs. conversion:

$$\overline{DP}_n = \frac{\text{concentration of consumed monomers}}{\text{concentration of chains}} = \frac{[M]_0 - [M]_t}{[P]} = \frac{[M]_0}{[P]} \cdot x_p$$

where $[P]$ denotes the *total* number of chains, active and inactive ones, which are generated in the transfer process.

In an **ideal polymerization** $[P] = [P^*] = f[I]_0$, where $[I]_0$ is the initial initiator concentration and f the initiator efficiency ($f = 1$ in most anionic systems, thus $[P] = [P^*] = [I]_0$).

If the propagating anions are free from transfer then the molecular weight of the chains corresponds to theoretical molecular weight depending on the monomer conversion.



In case of transfer, $[P]$ increases and the slope of the plot decreases.

Consequences of Termination and Transfer-Free Polymerization (4)

The **absence of termination and transfer** reactions has two important **consequences**:

1. The number-average molecular weight, M_n , of the resulting polymer is determined by the amount of consumed monomer and the initiator used for the polymerization;
2. All the chains at any time, t , propagate at the same rate and acquire the same length after a subsequent time interval, $t + dt$. This leads to a linear growth of polymer chains with respect to the monomer consumption, leading to a narrow distribution of chain lengths characterized by a *Poisson* distribution; the polydispersity index (PDI or $\bar{D}$) is given by:

$$\bar{D} = \text{PDI} = \frac{\overline{M}_w}{\overline{M}_n} \approx 1 + \frac{1}{\overline{\text{DP}}_n}$$

Consequences of Termination and Transfer-Free Polymerization (5)



It is important to note that **not all living polymerizations lead to narrow molecular weight distributions (MWDs).**

First, a *Poisson* distribution is obtained **only if the rate of initiation is much faster than that of propagation.**

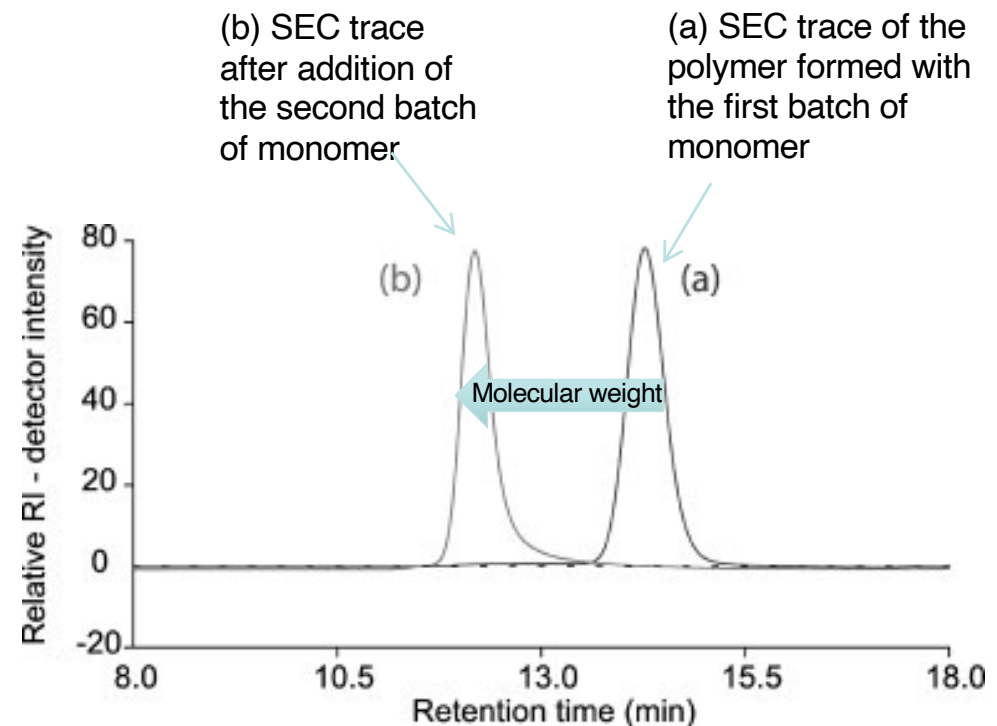
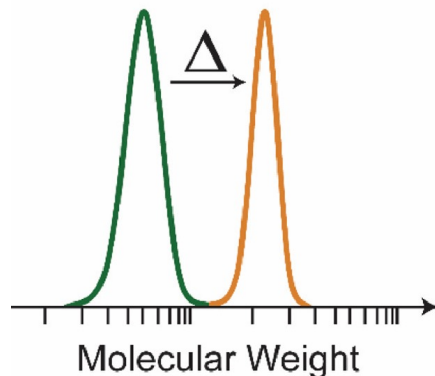
Second, many chain ends in anionic polymerizations (and also in other types of living polymerizations) can exist in various states, *e.g.*, covalent species, aggregates, various types of ion pairs, or free anions, which propagate at different rates or are inactive (“dormant”). Different types of chain ends in anionic polymerization exist in equilibrium with each other and a chain end can change its state depending on the reaction condition such as the polarity of solvent and the temperature:

- If the **rate of exchange** between these species is **rapid** compared to the rate of propagation, **the MWD remains narrow.**
- If the rate of exchange between these species is **slow** compared to the rate of propagation, this can lead to a **significant broadening of the MWD.**

Consequences of Termination and Transfer-Free Polymerization (6)

Monomer resumption experiment, also denoted as **chain extension experiment**, is another way to show the absence of termination using size exclusion chromatography (SEC) analysis.

A second batch of monomer is added after a certain period has elapsed after full monomer conversion.



In case of termination, one will find a bimodal distribution, one peak from the terminated chains and another from the active chains that participated in chain extension with the second batch of monomer.

Living polymer and living polymerization: Conclusions-Definitions*

The term **living polymer** was coined by Michael Szwarc to describe the products of the anionic polymerization of styrene initiated by electron transfer in tetrahydrofuran [1, 2].

In this context, “**living**” denotes the ability of a polymer chain to further add monomer after the initial batch of monomer has been consumed, and this means that the polymer chains do not undergo irreversible chain breaking reactions, such as termination or chain transfer.

The IUPAC Gold Book [3] defines “**living polymerization**” as **a chain polymerization from which chain transfer and chain termination are absent.**

It adds (although this is not part of the definition) the following: In many cases, the rate of chain initiation is fast compared with the rate of chain propagation, so that the number of kinetic- chain carriers is essentially constant throughout the polymerization. Typically, such a process should lead to a very narrow (Poisson) molecular weight distribution (MWD). However, a slow initiation process can have a considerable impact on the molecular weights achieved and on the MWD.

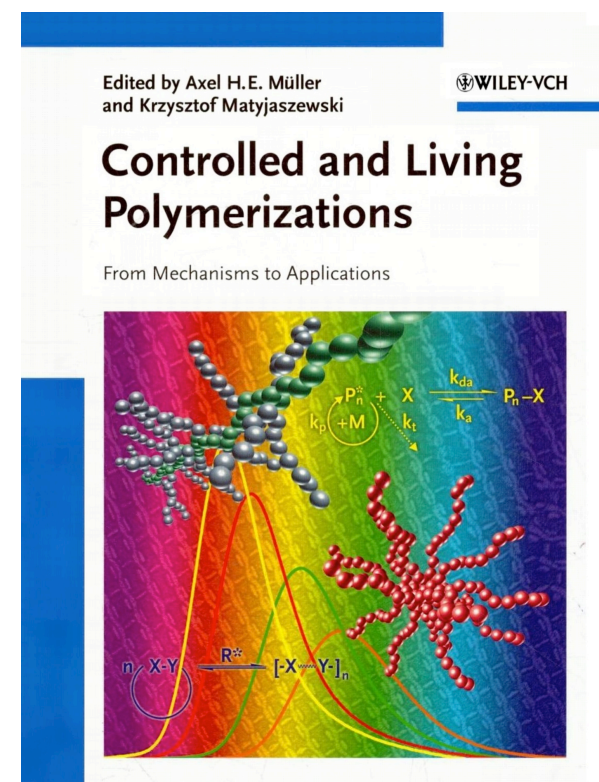
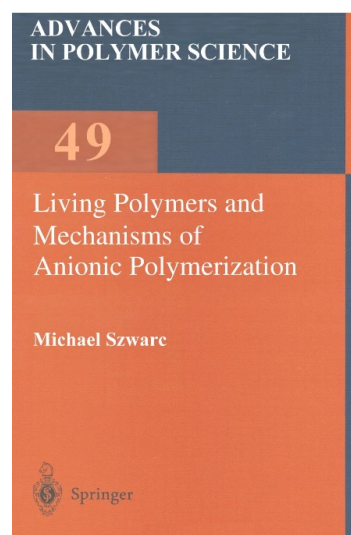
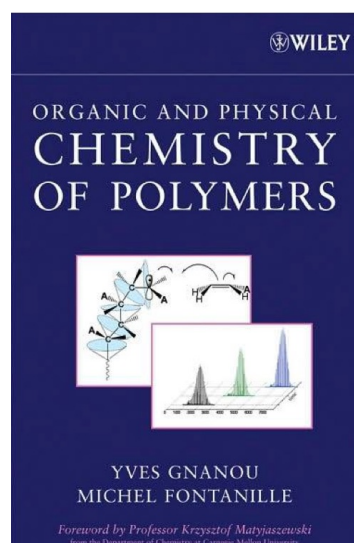
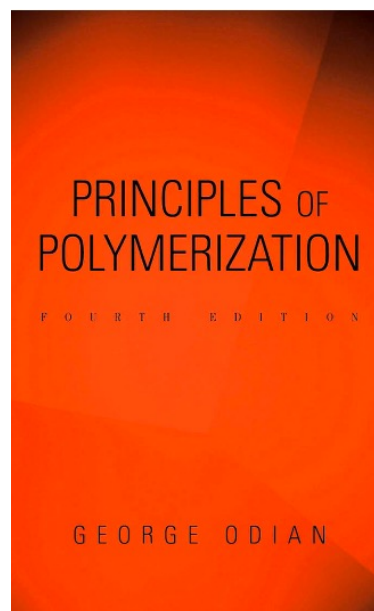
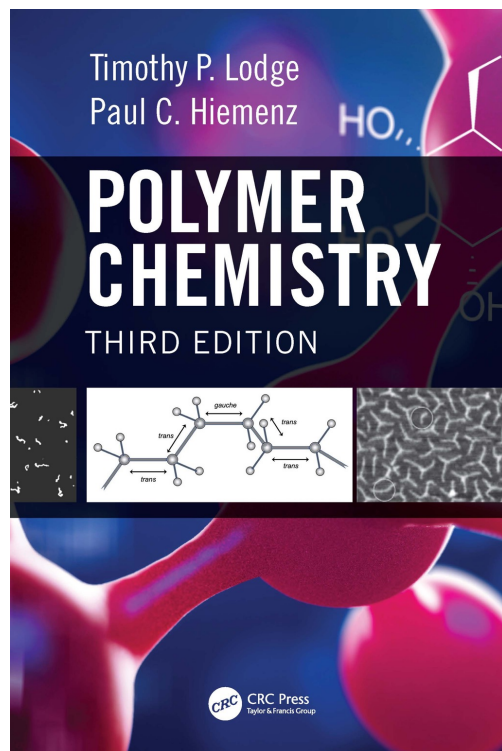
1. Szwarc, M., Levy, M. and Milkovich, R. (1956) J. Am. Chem. Soc., 78, 2656.
2. Szwarc, M. (1956) Nature, 176, 1168.
3. IUPAC Gold Book. <http://goldbook.iupac.org/L03597>.

* Axel H.E. Muller and Krzysztof Matyjaszewski (eds.), *Controlled and Living Polymerizations*, 2009 WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim

Anionic living polymerization - Summary

- Anionic polymerization is a form of **chain-growth polymerization** in which an anionic initiator transfers electrons (or negative charged group) to a monomer which then becomes the active center.
- The initiation step of an anionic polymerization involves a **nucleophilic attack** on a monomer resulting in an anion (carbanion, oxanion, ...). Basically, all vinyl monomers with (strong) electronegative substituents polymerize readily in the presence of these anions. Some electron-withdrawing substituents that stabilize the negative charge through charge delocalization, and hence permit stable anionic polymerization include -CN, -COOR, -C₆H₅, and -CH=CH₂, to name a few. Therefore, monomers such as styrenes, dienes, acrylates and methacrylates, epoxides, acrylonitriles and cyanoacrylates readily undergo anionic polymerization reactions.
- In carefully controlled systems (**pure reactants and inert solvents**), anionic polymerization does not undergo termination or transfer reactions. This type of polymerization is called **living polymerization**.
- Anionic living polymerization allows for **precisely controlled molecular weight**, and **nearly mono-disperse molecular weight distribution** (provided initiation is rapid relative to propagation).
- A number of important **applications** exist, but – except for rubber – not for the mass market, thermoplastic elastomers covering at least a niche market. This is partially due to the **necessity of intensive purification of all reagents** and low temperatures in some cases. In **industry**, anionic living polymerization is employed to produce synthetic **polydiene rubbers**, styrene-butadiene rubbers (**SBR**), and **thermoplastic styrenic elastomers**.
- When it comes to **very high molecular weights**, **very low dispersity** and very **precise block copolymer composition**, in particular for nanotechnology applications, anionic polymerization is still in the lead.

Bibliography



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*Tutorials
Problems 1-4*

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Tutorials – Problem 1

PROBLEM 1: ANIONIC LIVING POLYMERIZATION OF PROPYLENE OXIDE

The anionic polymerization of propylene oxide was carried out using the potassium salt of 1-methoxy-2-propanol as an initiator.



propylene oxide

1. Write the mechanism of the polymerization.
2. 0.1 g of initiator is used to polymerize 6 mL of monomer. What is the number average degree of polymerization expected if the conversion is complete ?
3. What quantity of initiator should be used to obtain a poly(propylene oxide) with a number average molar mass of $10,000 \text{ g.mol}^{-1}$ after 90% conversion? We start with 6 mL of monomer.
4. Anionic polymerization is called “living”. How is this type of polymerization characterized? What are the benefits of “living” polymerization ?

Data :

Propylene oxide. $d = 0.83$, $M = 58 \text{ g.mol}^{-1}$

Initiator: $M = 128 \text{ g.mol}^{-1}$

Tutorials – Problem 2

PROBLEM 2: ANIONIC LIVING POLYMERIZATION FOR THE SYNTHESIS OF A SURFACTANT

We want to prepare a nonionic polyether type surfactant. To do this, we first carry out the polymerization of propylene oxide (PO) using the potassium salt of dodecanol. The polymerization of propylene oxide is then followed by that of ethylene oxide (EO).

The initiator is prepared by reacting 2 mL of dodecanol dissolved in 8 mL of THF with 0.5 g of potassium. 5 mL of this solution are transferred to a flask containing 50 mL of dry THF, then 2.2 mL of propylene oxide are added slowly.

1. Write the reaction. What number average degree of polymerization will be obtained at full PO conversion ?
2. After complete reaction of the propylene oxide, 9 mL of ethylene oxide are added. What will be the structure of the copolymer finally obtained ? What number average degree of polymerization will be obtained at full EO conversion ? What is number average molecular weight of the final copolymer ?

Data:

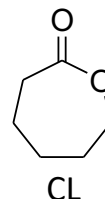
K: $M = 39 \text{ g.mol}^{-1}$. EO: $M = 44 \text{ g.mol}^{-1}$; $d = 0.88$. OP: $M = 58 \text{ g.mol}^{-1}$; $d = 0.83$.

Dodecanol: $M = 186 \text{ g.mol}^{-1}$; $d = 0.83$.

Tutorials – Problem 3

PROBLEM 3: APPLICATIONS OF ANIONIC LIVING POLYMERIZATION

The anionic polymerization of ϵ -caprolactone (CL) initiated with an aluminum-based initiator (I) is studied here. The following experimental conditions are employed: $[\text{CL}]_0 = 1.45 \text{ mol}\cdot\text{L}^{-1}$ and $[\text{A}]_0 = 0.14 \text{ mol}\cdot\text{L}^{-1}$.



The kinetic study of this polymerization is reported in Table 1.

1. Show that the kinetics follow an order 1 compared to CL.
2. Determine the apparent polymerization rate constant.
3. How long will it take to achieve 95% conversion?
4. Another study is devoted to the evolution of the number average degree of polymerization (DP_n) with the CL conversion p (Table 2). Discuss the values of DP_n .

Table 1. Kinetics of ϵ -caprolactone polymerization.

t (min)	0	5	30	100	130	180
$[\text{CL}(t)]/[\text{CL}(0)]$	1	0.95	0.82	0.55	0.45	0.33

Table 2. Evolution of the degree of polymerization in number DP_n .

Conversion p (%)	35	50	60	70	95
DP_n	5,4	6,8	8,5	9,8	12,8

Tutorials – Problem 4

PROBLEM 4: APPLICATIONS OF ANIONIC LIVING POLYMERIZATION

From an industrial viewpoint, living anionic polymerization plays an important role in producing several industrial materials, such as a styrene–butadiene rubber (SBR), and thermoplastic elastomers having both the elasticity and the processability, such as:

- Polystyrene-*block*-poly(1,3-butadiene)-*block*-polystyrene (PS-*b*-PB-*b*-PS or **SBS**),
- PS-*block*-polyisoprene-*block*-PS (PS-*b*-PI-*b*-PS or **SIS**), and
- the corresponding hydrogenated copolymers (**Kratons® G**).

1. Draw structural formulas (including various possible linkages) of **SBS** and **SIS** polymers.
2. Write out the general polymerization mechanism and the reactions sequences using appropriate initiators for the preparation of **SBS** and **SIS**.
3. Explain why **SBS** and **SIS** polymers are referred to as thermoplastic elastomers.
4. By hydrogenation of the **SBS** and **SIS** copolymers, the **Kraton® G** series are obtained. **SBS** copolymers lead to **SEBS** copolymers and **SIS** copolymers to **SEPS** copolymers (E for ethylene, P for propylene, B for butylene). Show that, starting from the structure of the central block of an **SBS** or **SIS** copolymer, one obtains by hydrogenation respectively **SEBS** or **SEPS** copolymers. Why are **Kraton® G** more resistant to UV and high temperatures than **SBS** and **SIS**? Why do they exhibit better compatibility with polyolefins?